Automated Waste Classification using Convolutional Neural Networks

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Abstract – Facilities that sort waste face many issues such as high contamination rates and other various inefficiencies due to manual sorting procedures. This paper presents an automated waste classification system that differentiates between different types of waste. Using a dataset of 2,467 total images with six trash categories, we use a Convolutional Neural Network built using pyTorch. We established a baseline of 23.67% accuracy based on assuming each object belongs to the most frequent classification. Our CNN optimized with batch normalization and dropout layers achieves an accuracy of 75%, showing a statistically significant improvement from the naive baseline.

Index Terms – Computer vision, image classification, CNN, convolutional neural networks, waste sorting.

I. Introduction

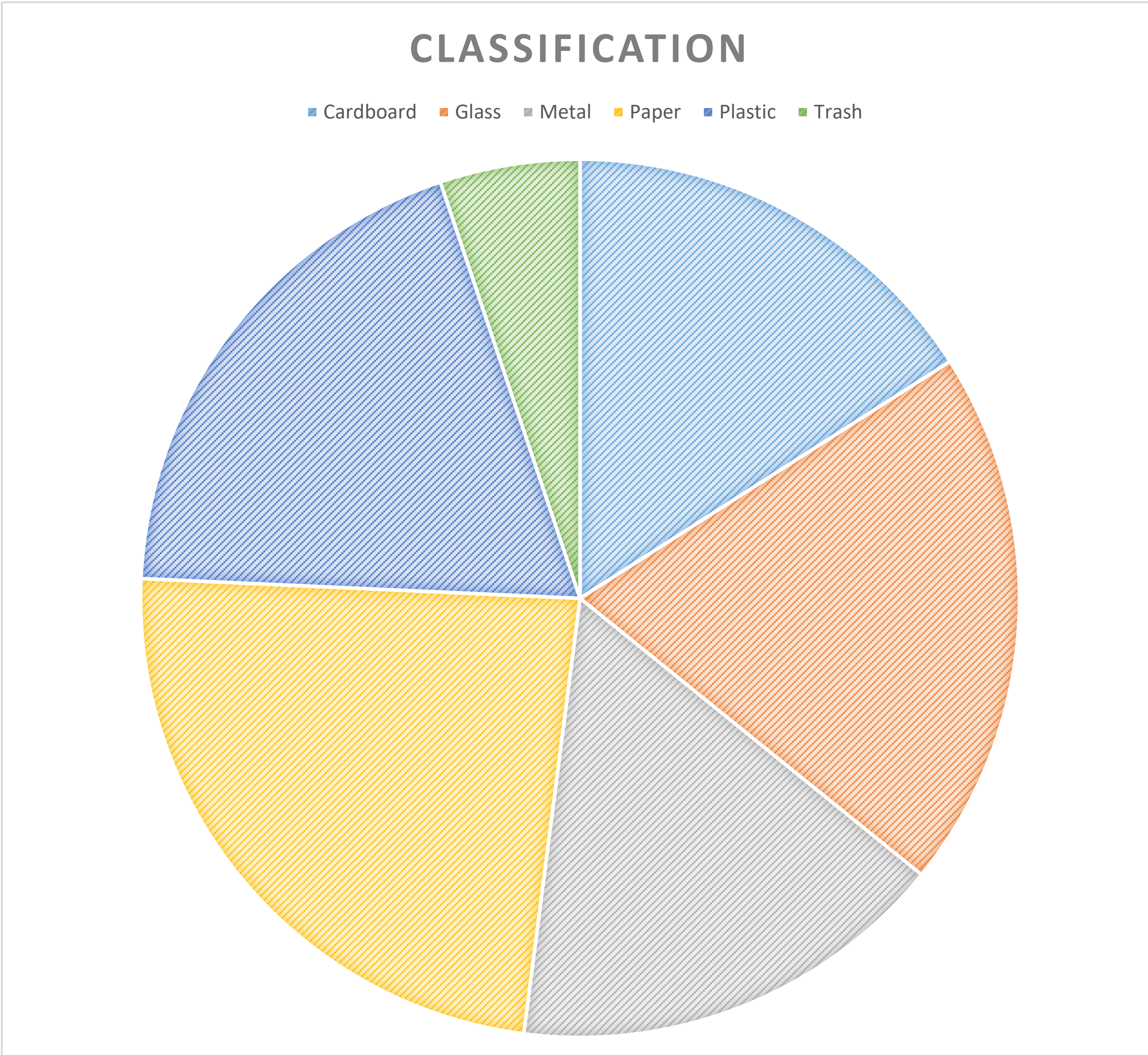
Recycling facilities receive lots of material and have to sort through it all manually to differentiate between trash and recyclables. This process is slow, inefficient, and causes high contamination rates. Global waste generation rates are only increasing and thus the need for the automated identification of these materials is needed more than ever.

This project aligns with the [CV] Computer Vision track for the CIS 530/730 term project. It also fits into Track 4, Perception and Agentic Systems. To address this ever-growing issue, we treat the process of waste sorting as an image classification problem with six total classes. We implemented a deep learning model for the purpose of outperforming a heuristic baseline model. Overall showing the potential benefit and viability of Convolutional Neural Networks for various environmental automation processes.

II. Methodology & Taxonomy

A. Dataset

The dataset used to train the data was from Kaggle. It contained 2,467 images with six separate image classifications [1]. The model was trained from this dataset and the categories and amount of images per category are as follows: cardboard (393), glass (491), metal (400), paper (584), plastic (472), and trash



(127). Before using the images, they were resized to the size of 128x128 and transformed into tensors.

B. Model Architecture

The system we proposed uses a Convolutional Neural Network made using PyTorch. There are 3 layers with paddings of 1 and kernel sizes of 3x3. Each convolutional layer is followed by both a Batch Normalization layer and a Rectified Linear Unit function, BatchNorm2d and ReLU. The data is flattened and passed into a dropout layer with p=.5 to minimize overfitting. The final output layer produces one of six target classes.

III. Baseline

In order to prove that the advanced deep learning Convolutional Neural Network was efficient. We established a naive baseline. We developed this baseline by simply identifying the most common class in the dataset, which is paper, with 584 images. In testing, we assume all images fit into this waste classification. This yields an accuracy of 23.67% when testing the entire dataset.

FIGURE 1: CONVOLUTIONAL NEURAL NETWORK ARCHITECTURE. THREE CNN BLOCKS WITH NORMALIZATION FOLLOWED BY DROPOUT LAYERS INTO FINAL SIX CLASS OUTPUT LAYER.

IV. Experiments and Rigorous Evaluation

A. Model Training

This model was trained over 15 epochs. We used cross-entropy loss and an Adam optimizer with a .001 learning rate. Before passing in the data, we applied slight data randomization such as flipping and rotation. This was done to expand the dataset size and prevent potential overfitting. This model achieved a training accuracy of ~68%.

B. Statistical Significance and Evaluation

To ensure that the Convolutional Neural Network's performance was valid. We did a McNemar's test. The test was applied to the outputs for both the baseline and the CNN results. It yielded a p-value of 6.68E-66 which is significantly smaller than the threshold of $< .001$, meaning that the CNN shows a statistically significant improvement over the baseline.

V. Future Work & Challenges & Discussion

The biggest problem/challenge with this project was trying to minimize overfitting. The smaller size of the dataset, 2,467 images. A valid solution that definitely helped was implementing batch normalization. For future work, a big priority would be the expansion of this dataset. It could be both bigger and more balanced since certain classifications are significantly smaller and have way less training data. This would increase training accuracy and reduce bias towards the larger classifications of data.

REFERENCES

[1] "Garbage Classification" www.kaggle.com. https://www.kaggle.com/datasets/asdasdasasdas/garbage-classification